



What is the API Gateway Pattern ?

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API - It stands for "Application Programming Interface". In general APIs are built-in libraries or collections of libraries that perform some specific task or function. In general, we use API to get connected with other web applications. We generally connect with other web applications by following certain protocols, methods. We can also use the data through the API that is used by another web application or software through the use of API of that software. In other words, we can exchange our data through API with other web applications/software.

What is API Gateway Pattern: The API Gateway Pattern in some cases stands for "Backend for frontend". It basically the entry gate for taking entry in any application by an external source. The pattern which is going on in a programmer's mind while they are making the client's application. It acts as a medium between the client applications and microservices. For example-Netflix is the most famous example of an API gateway.

Use of the API Gateway Pattern: The main responsibility of this pattern is that it routes the request means basically provide a road map for how our request goes, approve or may be canceled, API composition, and app authentication.

How API Gateway Handles the Client requests:

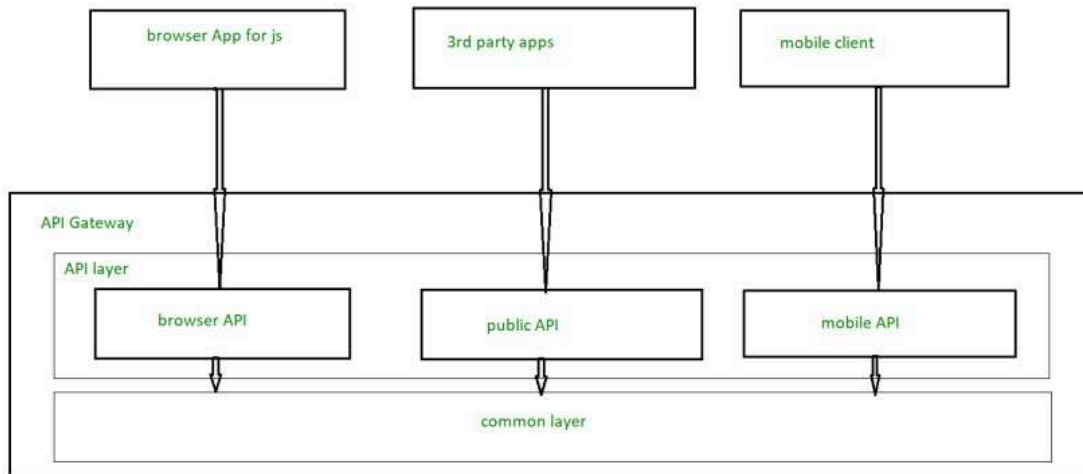
- API requests mostly come from clients who are external and firstly go to API gateway.
- It is the working of API gateway which routes the request to their place.
- The other additive requests i.e. using multiple services and the aggregating result are handled by API gateway.

It helps in preventing our software from malicious cyber fault with its two different layers one is a common layer that is used for authentication among the client-side and the API layer. The layers secure us from attacks like XML Parser exploits, SQL injection, etc. It helps clients to get communicate very easily.

The architecture of API gateway - It basically consists of two layers -

- A common layer helps in the working of edge function which helps in the authentication.
- In the API layer, each API module helps in making an API for specific clients. API layers consist of one or more independent API modules.





Advantages of API gateway pattern -

- It encloses the whole internal structure of web applications.
- It never calls a particular service. For example, client interaction with API gateway.
- It helps in the simplification of code of the client-side.

Disadvantages of API gateway pattern -

- It is an important component for every web application means the web application services will be shown only if the API is up-to-date means updated.
- It becomes very important for each process for being lightweight because otherwise their time complexity will get increased because their developer has to wait in the process of updating API.

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